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Ground-based research to develop the science and technology of microfluidic and dielectric movement and attachment of DNA onto surfaces for subsequent DNA processing and analysis. This will follow up on NASA grant NAG3-2134 by using the flow fields studied in that grant to stretch and deposit double stranded DNA onto treated substrates, and to study the hydrodynamic and physico-chemical forces controlling the stretching and deposition processes. The flows to be used are: (1) a simple shearing flow produced by a rotating parallel disk parallel to a stationary disk, and (2) the flow in an evaporating droplet. In addition to these, fundamental studies will be carried out on dielectrically-driven stretching and movement of DNA molecules between electrodes. We will study DNA flow and deposition in (1) a simple shearing flow produced by a rotating parallel disk parallel to a stationary disk, and (2) the flow in an evaporating droplet. Finally, (3) we will determine the mechanism of dielectrically-driven stretching and movement of DNA molecules between electrodes.
Using a high-resolution CCD camera, we will measure quantitatively the stretching and anchoring of DNA on substrates

Task Description:	in these two flows. We will develop a model for the stretching of DNA near surfaces and for the adhesive forces exerted by the surfaces on the DNA. By comparing detailed distributions of molecular stretch of DNA adsorbed onto the surfaces with Brownian dynamics simulations of these distributions, we will validate the DNA deposition and adhesion model. We will also explore the use of confocal microscopy for quantitative measurement of the droplet-drying flow field. Stretching and anchoring DNA molecules is an important first step to studying DNA/protein interactions, including repair of double-strand breaks caused by ionizing radiation in space. The deposition of DNA using micro-flows is also interest in genomics and micro-arrays and so has wide applications. The addition of electric field effects provides an alternative means of stretching DNA molecules while retaining them in solution for restriction digestion and other applications. Stretching and anchoring DNA molecules is an important first step to studying DNA/protein interactions, including repair of double-strand breaks caused by ionizing radiation in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Advanced methods of manipulating, stretching, aligning, and depositing DNA molecules will further study of DNA interactions with proteins, which is key to packaging, transcription, and repair of DNA, all vital processes with close connections to disease states, including cancer.
Task Progress:	Single-molecule studies are increasingly used to investigate the dynamics and statistics of biochemical and biophysical interactions between DNA and proteins that that are difficult to discern in bulk experiments that average over an ensemble of molecules. Recent advancements in fluorescence imaging tools such as total internal reflection fluorescence (TIRF) microscopy and cooled charge-coupled-device cameras have made visualization of single biomolecules possible. Such visualization studies allow imaging of DNA transcription, DNA repair, and other interactions between proteins and DNA molecules that are at the heart of diseases such as cancer, that can be induced by radiation, including solar radiation that puts astronauts at risk during prolonged space flight. Single DNA molecules can be stretched and manipulated by flow, electric, magnetic, and optical fields. Of these, the use of flow fields is probably the most convenient because it does not rely on complex tools such as optical or magnetic tweezers or high-frequency electric fields. There are two types of stretching force exerted on DNA molecules in flows, namely viscous drag forces produced by the bulk flow, and meniscus forces generated by motion of an air-water interface along the DNA molecule. We have compared the efficiency and effectiveness of DNA stretching and deposition by “molecular combing,” spin-stretching, air-blowing, and drying droplet flows, all of which use some combination of bulk flow and meniscus motion. We have also used simple shearing flow, generated by a rotating disk, and imaged DNA stretching in situ. Flows with Moving Meniscus In molecular combing, a substrate with a hydrophobic surface is dipped into a DNA-containing solutions, and as the substrate is pulled out from the solution the DNA sticks at one end to the surface and is stretched out by the force exerted on the rest of the DNA molecule by the receding meniscus, before the other end also sticks to the surface. In spin-stretching, previous studies suggest that DNA molecules are stretched by the centrifugal force exerted on a droplet of DNA solution that is proportional to the angular velocity of the disk and the radial distance of the DNA from the center of rotation. It was shown previously that DNA molecules could be spin-stretched onto a mica surface or a cover glass coated with MgCl2 or CaCl2.. However, careful analysis of the flow were not conducted in these early studies, and so the mechanism of stretching was not demonstrated. By blowing an air jet at the side of a droplet of DNA solution placed on a hydrophobic substrate, droplet motion and droplet flows are generated that stretch DNA molecules as they are deposited onto the substrate. When DNA molecules are air-blown onto a polydimethylsiloxane (PDMS) surface, the stretched molecules can be subsequently transferred onto a non-treated cover glass or mica surface through micro-contact printing. Our group and others have also stretched DNA molecules in a flow generated by drying a droplet deposited onto a positively charged surface such as one treated with aminopropyltriethoxy silane- (APTES). When the contact line is pinned to this surface, a radial flow is created to replenish fluid that is evaporated from the periphery of the drop. This flow both transports DNA molecules towards the periphery and stretches them as they are attached to the substrate. The goal of our work is to help develop the engineering science required to optimize methods of single-molecule DNA processing, which will be important in the study of DNA-protein interactions and to portable monitoring of DNA-protein interactions, such as those involved in DNA repair. We have compared the degree of stretching and adsorption of DNA molecules onto substrates by molecular combing, droplet drying, spin-stretching, and air-blowing. With molecular combing and spin-stretching, we find that DNA molecules stretch to a greater degree on a PMMA surface at pH 8.0 than on either a PMMA surface at lower pH or a PS surface at the same pH, in contrast to the previously held idea that DNA molecules stretch more effectively when they have hydrophobic interactions with the substrate achieved by lowering the pH or increasing the hydrophobicity of the substrate. The stretching is apparently enhanced at pH 8.0 because at this pH the DNA bases are not protonated and therefore not locally melted; the DNA then adsorbs primarily at its ends, reducing the chance for attachment along the DNA backbone to provide anchor points to support the force that would otherwise be transmitted to the DNA. This DNA stretching at pH 8.0 can further be enhanced on PMMA surfaces which have reduced nonspecific interactions with the DNA relative to PS surfaces. For the same reason, the number density of DNA molecules decreases on PMMA relative to PS surfaces. We stretched lambda phage DNA and T2 DNA molecules onto APTES-treated glass surfaces via drying droplet flow. Our results for the mean stretch ratio agreed well with that from the Brownian dynamics simulations and with previously measured stretch near the edge of the droplet by Chopra et al. We find that T2 DNA molecules near the periphery of the droplet stretch less because they tend to stick to the surface sooner than those closer to the center and therefore experience a smaller shear gradient at the time they adsorb. We find by high-speed video microscopy, that there is complex droplet deformation and motion during air-blowing, which complicates the deposition and stretching process, leading to radial deposition patterns. However, over-stretching is achieved in air blowing, as it is in spinning and molecular combing because all these methods induce a moving contact line, which seems to be necessary to generate the force needed for overstretching. Only in droplet drying is the deposition almost entirely due to the flow within the fluid, and this method results in no overstretching and rather poor stretching overall. In the methods that do produce a moving contact line, stretching is optimized at pH 8.0, presumably because at this physiological pH, local DNA melting due either to protonation of bases (at low pH) or deprotonation of phosphate group (at high pH) is minimized. In general, we have shown that, by varying the stretching method, the surface, and the solution pH systematically, using controlled solution compositions, we can see systematic trends that can be given reasonable interpretations and provide a basis for choosing and optimizing a method for a desired degree of stretch and deposition density. This should be helpful in using these methods for single-molecule studies, for example of DNA-protein interactions. However, there are numerous unexplained observations, including the minimum in DNA stretching in the spin-stretching method that we observe in

	the air-blowing method on polystyrene at pH 6.6, and the sensitivity of the results to the particular surface. The very complex interfacial movements observed by high-speed photography of the spin-stretching and air-blowing processes make interpretation of the stretching and deposition achieved in these flows very difficult. Future work, using molecular dynamics simulations to examine the local interactions of DNA with various surfaces, fluid mechanical calculations to determine the flow near moving interfaces, and Brownian dynamics simulations to compute DNA configurations in these flows, are recommended to gain deeper understanding of DNA stretching and deposition methods. Simple Shearing Flow We have also imaged DNA molecules during a simple shearing flow in a torsional parallel disk shearing device. The device uses a transparent glass lower disk and epifluorescence microscopy to directly image the DNA molecules undergoing shear. From these experiments, we have discovered that the DNA molecules are driven to migrate away from the surface during shearing flow. This affects the ability of the DNA to stretch and successfully anchor to the surface in flows that contain only shearing, and lack a component that can drive the molecule to the surface. We also quantitatively compared the measurements with Brownian dynamics simulations. Imaging Drying Droplet Flow We have also been imaging the flow in an evaporating droplet, which is a simple flow used to stretch and deposit DNA onto surfaces, in preparation for study of DNA/protein interactions. While we have predicted that there should be large Marangoni stresses in the drying of a water droplet, we could hardly observe any Marangoni flow in such droplets, probably because of the difficulty in obtaining clean water surfaces, unaffected by traces of surface-active agents. So, to confirm the effect of Marangoni stresses in drying droplets, we studied the flow in a drying octane droplet, whose surface remains clean for much longer periods. To study this flow conveniently, we have developed an imaging method using a micro laser sheet, and a method of synthesizing fluorescent particles that are stable in octane, and of sufficiently large size to be viewed as flow tracers in the laser sheet. The synthesis of these polymethyl methacrylate (PMMA) particles constitutes an unexpected breakthrough that transcended the original goals of the project. Using these particles, we successfully imaged a strong Marangoni flow in a sessile droplet, for the first time. In addition, to correct for the effects of refraction of fluorescent light as it emerges from the droplet, we developed a ray-tracing algorithm, that allows us to compare quantitatively the predicted, with the observed, particle recirculation patterns in the micro-flow in the droplet.
Bibliography Type:	Description: (Last Updated: 02/25/2006)
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Abstracts for Journals and Proceedings	Hu, H.; Larson, R.G. "Analysis of the Effects of Marangoni Stresses on the Micro-Flow in an Evaporating Sessile Droplet" Langmuir 21:3972-3980 , Apr-2005